# AP Integration Specification for NMS Control and Bundle Deployment

## 1. Purpose

This document specifies the required behavior for Access Points (APs) that integrate with the Network Management System (NMS). The design mirrors the robot bundle architecture already implemented in the system, but with a simplified runtime model appropriate for AP hardware. The objective is to allow the NMS to remotely manage AP state, receive periodic status and traffic reports, and deploy bundle updates in a consistent and recoverable way.

## 2. Runtime Architecture Overview

The AP runtime model contains three main functional components:

- **Registration component** – ensures the AP registers itself with the NMS.
- **Agent component** – polls the NMS for commands every 10 seconds.
- **Uploader component** – periodically sends status and traffic information to the NMS.

Unlike the robot system, the AP does not require GUI, voice modules, or scanning subsystems. The AP runtime environment therefore remains lightweight.

## 3. First-Time Startup Procedure

When an AP is installed for the first time, the following sequence must be executed locally:

1. Run AP registration until success.
2. Start the AP command polling agent.
3. Start the AP uploader process.

### Example startup commands:

```
bash /opt/_RunScanner/ap_register_until_ok.sh
python3 /opt/_RunScanner/ap_agent.py
python3 /opt/_RunScanner/ap_uploader.py
```

If a service manager is available (systemd or OpenWrt init), these commands should be encapsulated into persistent services that start automatically at boot.

## 4. Bundle Deployment Workflow

Bundle deployment follows the same model used for robots. The process ensures that updates are delivered safely and that the AP always returns to a known operational state.

1. Build bundle using ap_make_bundle.sh
2. Upload bundle ZIP file to the NMS bundle repository

3. First-time installation: AP directly downloads bundle using GET /bootstrap/bundle/{bundle_id}
4. AP extracts and applies the bundle using install.sh
5. AP reboots automatically after installation
6. AP registers with the NMS again after reboot
7. Subsequent updates use the NMS command queue with bundle.apply

# 5. Command Polling Model

The AP agent polls the NMS every 10 seconds. Each poll retrieves pending commands assigned to the AP. The agent then dispatches the command to the appropriate handler function.

Each command execution returns two values:

- status – 'ok' or 'error'
- detail – diagnostic string returned to the NMS

# 6. Required Callback Functions

The AP implementation must provide several callback functions that interact with hardware or firmware features. These callbacks allow the generic NMS control framework to operate on different AP platforms.

| Function | Purpose |
|---|---|
| ap_get_status_snapshot() | Return current AP state including MAC, IP, channel, and association tab |
| ap_get_traffic_snapshot() | Return per■STA traffic statistics and frame duration information. |
| ap_cmd_get_association() | Return association table of connected stations. |
| ap_cmd_sta_associate(sta_mac) | Force association of a specific station. |
| ap_cmd_sta_disassociate(sta_mac) | Disconnect a station from the AP. |
| ap_cmd_set_sta_txpower(sta_mac, txpower) | Set transmit power for a specific station. |
| ap_cmd_set_ap_txpower(txpower) | Adjust AP global transmit power. |
| ap_cmd_traffic_enable() | Enable periodic traffic reporting. |
| ap_cmd_traffic_disable() | Disable periodic traffic reporting. |

# 7. Expected Data from Status Snapshot

- Timestamp
- AP name
- IP address
- MAC address
- Operating band (2.4/5/6 GHz)
- Channel number
- Number of antennas

- Station association list

# 8. Expected Data from Traffic Snapshot

- Per-station statistics
- Frame count
- Average frame duration
- MCS distribution
- Per access category traffic

# 9. Operational Differences Between Robot and AP

| Feature | Robot | AP |
|---|---|---|
| GUI / Autostart | Yes | No |
| Voice system | Yes | No |
| Wi■Fi scanning subsystem | Yes | No |
| Core services | agent, uploader, poller, voice, avstream | agent, uploader only |
| Hardware callbacks | Robot motion + sensors | AP radio control functions |

# 10. Summary

The AP integration is designed to reuse the same NMS command and bundle infrastructure used by robot devices. Only the hardware■specific callback layer differs. This allows heterogeneous devices (robots and APs) to coexist under a unified NMS architecture.